

GATE

ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 10



By- Amulya Ratan Sir

Recap of Previous Lecture



Topic

Number System



Topics to be Covered



Topic-1

Counting Theory

Topic-2

Time and Work



[MCQ]



#Q. Which largest number of 5 digits is divisible by 99?

Assignment

$$99 = \underline{9} \times \underline{11}$$

~~A~~

$$\underline{99999} \begin{array}{r} 27 \\ 18 \\ \hline 9 \end{array} = 45$$

~~B~~

$$\underline{99981} \begin{array}{r} 19 \\ 17 \\ \hline 2 \end{array} = 36$$

~~C~~

$$\underline{99909} \begin{array}{r} 27 \\ 9 \\ \hline 18 \end{array} = 36$$

D

$$\underline{99990} \begin{array}{r} 18 \\ 18 \\ \hline 18 \end{array} = 36$$

[MCQ]



#Q. Find the least perfect cube divisible by 2, 3, 4 and 6.

Assignment

A

216

= 9

B

1728

= 18

C

512

= 8

D

360

[MCQ]



#Q. How many natural numbers up to 100 are divisible by both 2 and 3?

Assignment

$$6 \times \underline{1} = \underline{6}$$

$$\underline{6}$$

A

13

B

14

C

17

D

16

$$\underline{16}$$

$$6 \times \underline{16} = \underline{96}$$

[MCQ]



#Q. Find the largest four-digit number that is exactly divisible by 88.

Assignment

$$= 8 \times 11$$

☒ A

$$\begin{array}{r} 13 \\ 9844 \\ \underline{13} \\ 12 \\ \underline{12} \\ 0 \end{array}$$

☐ B

$$\begin{array}{r} 13 \\ 9768 \\ \underline{13} \\ 15 \\ \underline{15} \\ 0 \end{array}$$

☒ C

$$\begin{array}{r} 13 \\ 8894 \\ \underline{13} \\ 17 \\ \underline{17} \\ 0 \end{array}$$

☒ D

$$\begin{array}{r} 13 \\ 9944 \\ \underline{13} \\ 13 \\ \underline{13} \\ 0 \end{array}$$

L^y
(+3)

901 - 2000

(11) ~~N-1~~

1st Jan

How many times
SUNDAY

11 + 3 = 14

1st Jan, 1901 → ? Tuesday

- 1901 → Tue
- 1902 → Wed +1
- 1903 → Thu +1
- 1904 → Fri +1
- (1905) → Sun +2
- (+11) → 1989 ✓
- (+6) → 1995 ✓

- 1900 → |
- 2 →
- (+11) 1950 ✓
 - (+11) 1961 ✓
 - (+6) 1967 ✓
 - (+11) 1978 ✓

- ✓ 1905
- (+6) ✓ 1911
- (+11) ✓ 1922
- (+11) ✓ 1933
- (+6) ✓ 1939

1928 ✓

1956 ✓

1984 ✓

✓ 1905 → Sunday

1906 → Mon +1

1907 → Tue +1

1908 → Wed +1

1909 → Fri +2

1910 → Sat +1

✓ 1911 → Sun +1

1912 → Mon +1

1913 → Wed +2

1914 → Thu +1

1915 → Fri +1

1916 → Sat +1

1917 → Mon +2

1918 → Tue +1

1919 → Wed +1

1920 → Thu +1

1921 → Sat +2

1922 → Sun +1

1923 → Mon +1

1924 → Tue +1

1925 → Thu +2

1926 → Fri +1

1927 → Sat +1

1928 → Sun +1

Counting Theory

Numbers

What?

* the number of ways a work can be done

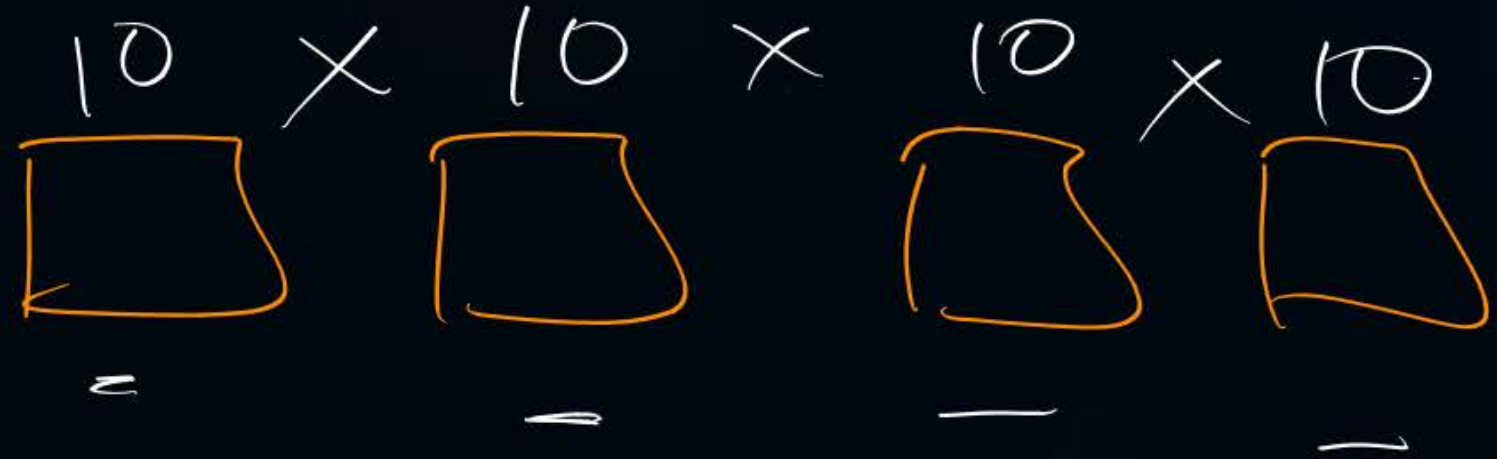
OR (+)

AND (x)

[MCQ]



#Q. In how many different ways you can choose your ATM PIN ?



6
7
8
9

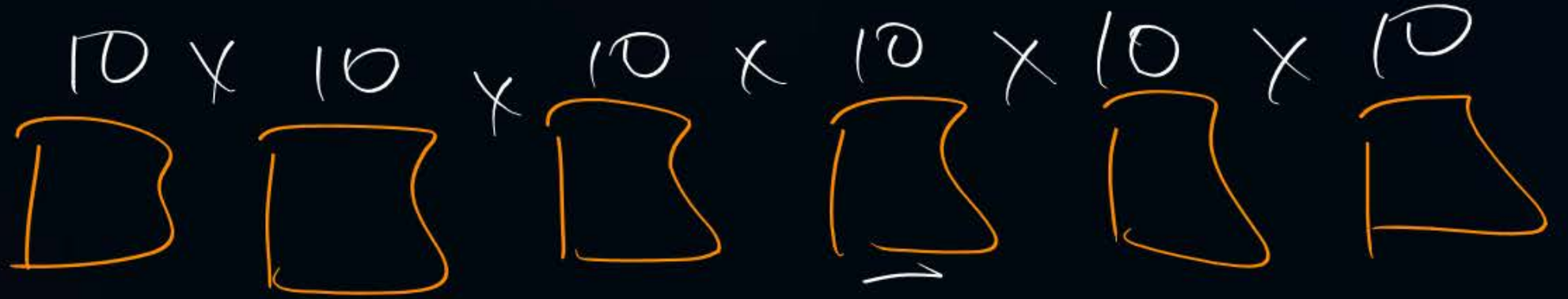
0
1
2
3
4
5

10,000

[MCQ]



#Q. In how many different ways you can select a six digit phone password/passcode ?



0-9

$$= 10 \times 10 \times 10 \times 10 \times 10 \times 10$$

[MCQ]



#Q. How many four [✓]digits number can be formed using 1, 2, 3, 4 & 5?

~~Repetition
Allowed~~

$$\begin{array}{c} \text{Th} \\ \boxed{5} \end{array} \times \begin{array}{c} \text{H} \\ \boxed{5} \end{array} \times \begin{array}{c} \text{Ten} \\ \boxed{5} \end{array} \times \begin{array}{c} \text{Unit} \\ \boxed{5} \end{array}$$

$$= \frac{625}{1}$$

~~Per~~

$$\begin{array}{c} \text{Th} \\ \boxed{5} \end{array} \begin{array}{c} \text{H} \\ \boxed{4} \end{array} \begin{array}{c} \text{Ten} \\ \boxed{3} \end{array} \begin{array}{c} \text{Unit} \\ \boxed{2} \end{array}$$

$$5 \times 4 \times 3 \times 2$$

$$(5-1) (5-2) (5-3)$$

$$= 120$$

[MCQ]



#Q. In how many different ways you can choose your ATM PIN which is an even ?

10 x 10 x 10 x 5

□ □ □ □

0-9

0-9

0-9

0

2

4

6

8

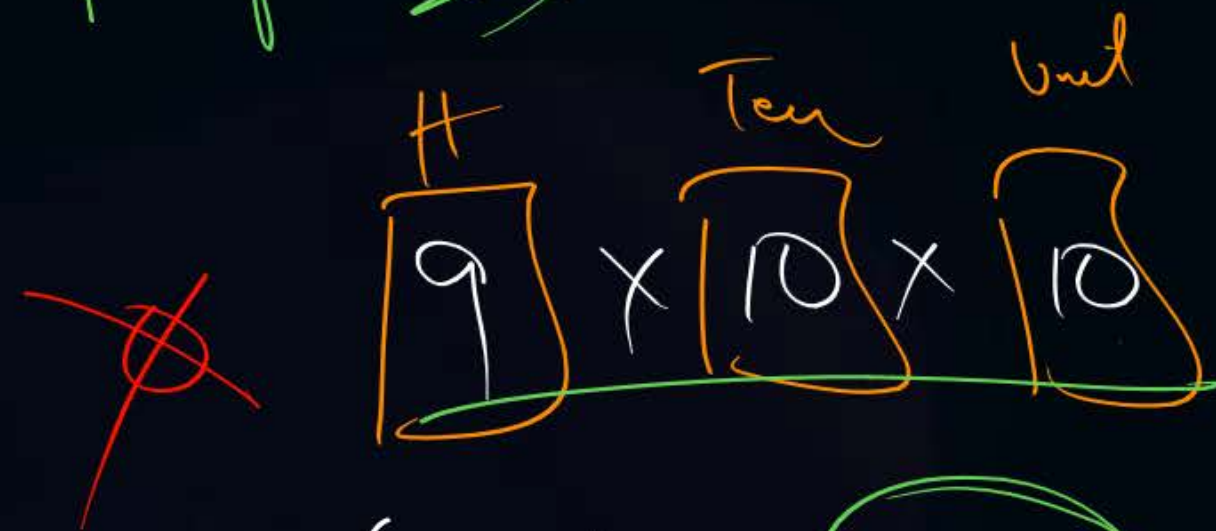
= 5000

[MCQ]



#Q. How many three digits number can be formed using digits 0 to 9?

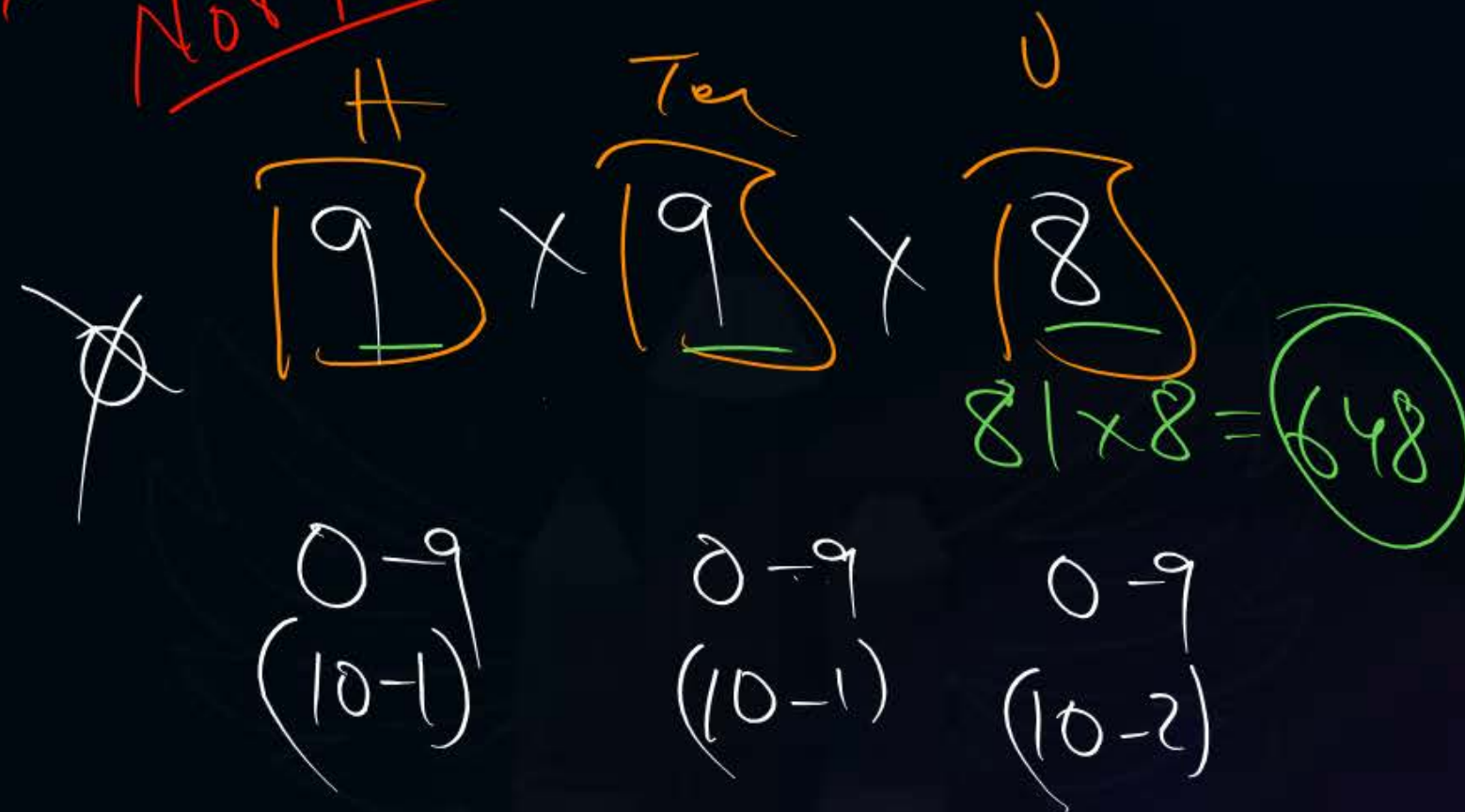
Rep ALLOWED



$(10-1)$

900

~~Rep~~ Not Allowed



[MCQ]



#Q. In how many ways four letters can be posted in 6 post boxes, if each box can take any number letters?

$$\begin{aligned} &L_1 \quad L_2 \quad L_3 \quad L_4 \\ &6 \times 6 \times 6 \times 6 \\ &= 36 \times 36 = 1296 \end{aligned}$$

A 4096 ?

B 1500

C 24

D 1296

[MCQ]



#Q. In how many ways can you arrange 5 Novels on a shelf?

A 720

B 120

C 360

D 300

$$5 \times 4 \times 3 \times 2 \times 1 = 120$$

N_1
OR
 N_2
OR
 N_3
OR
 N_4

$(5-1)$ $(5-2)$

OR
 N_5

[MCQ]



#Q. Four persons A, B, C and D are to be seated in a row. C should not be seated at the second position from the left end of the row. The number of distinct seating arrangements possible is?

Assignment

A

6

B

9

C

18

D

24

[MCQ]



#Q. Five different book (P, Q, R, S, T) are to be arranged on a shelf. The books R and S are to be arranged first and second, respectively from the right side of the shelf. The number of different orders in which P, Q and T may be arranged is?

Assignment

A

12

B

120

C

6

D

2

[MCQ]



#Q. A license plate begins with 3 letters. If the possible letters are A, B, C, D and E, how many different ways these letters can be written if no letter is used more than once?

Assignment

- A** 720
- B** 120
- C** 60
- D** 30

[MCQ]



#Q. In how many different ways can the letters of the word 'PLEADING' be arranged in such a way that the vowels always come together?

Assignment

- A** 720
- B** 4320
- C** 5040
- D** 3600

Direct & Inverse:

$$\left[\begin{array}{c|c} + & \\ \hline \times & \\ \hline \cdot & \end{array} \right]$$

$$\left[\begin{array}{c|c} \cancel{+} & \cancel{-} \\ \hline \cdot & \times \end{array} \right]$$

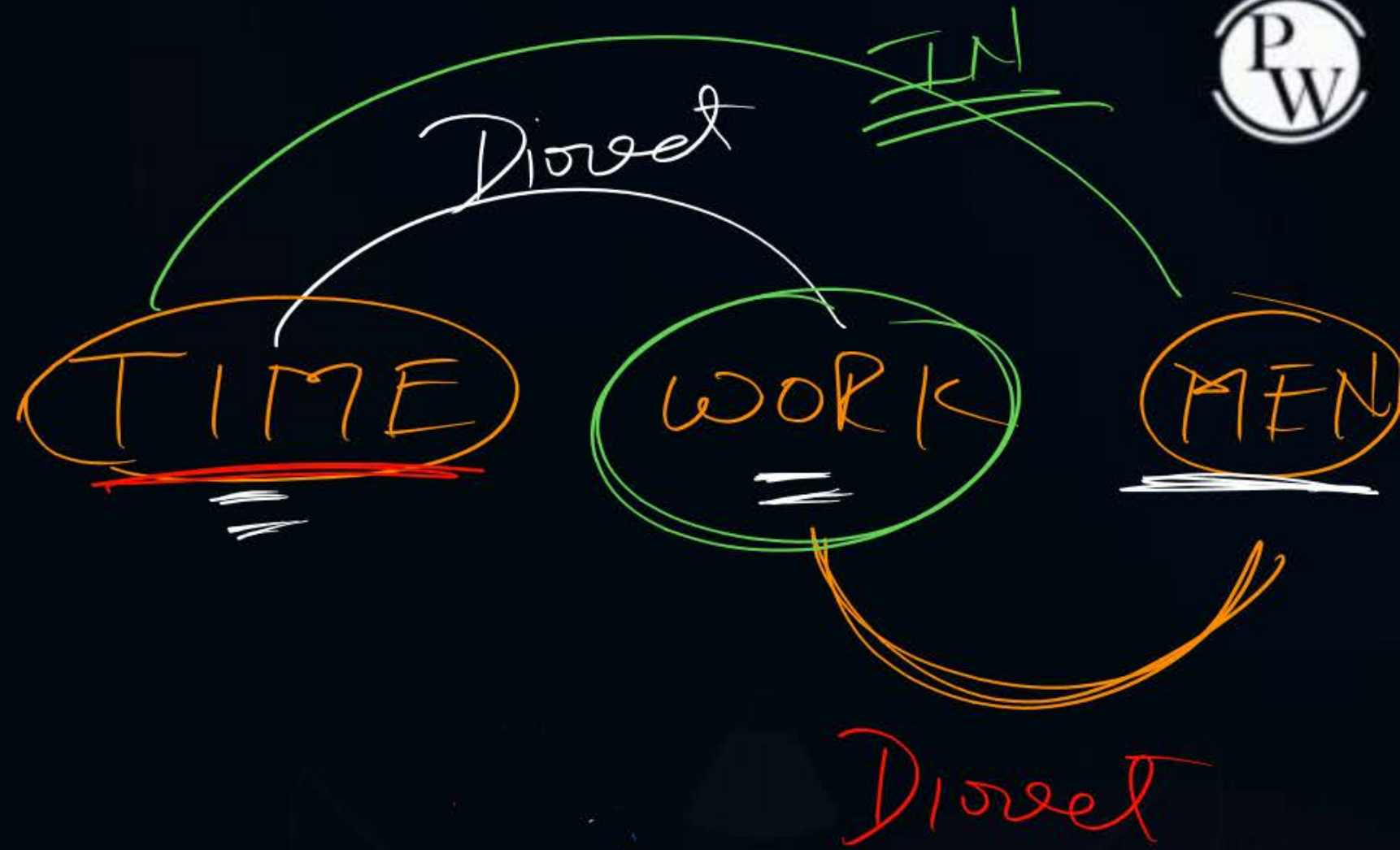
Time & Work

$\frac{1}{10}$ $A = 10$ days
 $\frac{1}{20}$ $B = 20$ days

$A \& B = \frac{1}{10} + \frac{1}{20} = \frac{3}{20}$



$\frac{20}{3}$ OR $6\frac{2}{3}$ days OR 6.6 days



Let's Try:



Q.

$$A = 20$$

$$B = 30$$

12 days

Q.

$$A = 25$$

$$B = 40$$

$15\frac{5}{13}$ days

Q.

$$A \& B \Rightarrow 40$$

$$A \Rightarrow 60$$

$$B \Rightarrow ?$$

$$120$$

[MCQ]



#Q. A can do a work in 10 days whereas B in 20 days. They started working on alternate days, i.e. 1st day 'A', 2nd day 'B', 3rd day 'A'.....so on. Then in how many days the total work would be completed ?

Handwritten solution:

$A = \frac{1}{10}$ | $B = \frac{1}{20}$

$\frac{1}{10} + \frac{1}{20} = \frac{3}{20} \Rightarrow 2 \text{ days work}$

$\frac{18}{20} \Rightarrow 12 \text{ days}$

13 days ✓

[MCQ]

① B ② A ③ B ④ A - - - - -



#Q. A can do a work in 10 days whereas B in 20 days. They started working on alternate days, i.e. 1st day 'B', 2nd day 'A', 3rd day 'B'.....so on. Then in how many days the total work would be completed?

$$\frac{1}{20} = \frac{1}{10} ?$$

$$A = \frac{1}{10} \quad | \quad B = \frac{1}{20}$$

$$\frac{1}{20} + \frac{1}{10} = \frac{3}{20} \xrightarrow{\times 6} 2 \text{ days}$$

1 work A = 10
 $\frac{1}{20}$ " A = $10 \times \frac{1}{20}$
 $\times 6$
 $= \frac{1}{2}$

13th day

$$\frac{1}{10} - \frac{1}{20} = \frac{1}{20} ?$$

$$\frac{18}{20} \rightarrow \underline{\underline{12 \text{ days}}}$$

13½ days
 OR
13.5 days

[MCQ]



#Q. A can do a work in 25 days whereas B in 40 days. They started working on alternate days, i.e. 1st day 'A', 2nd day 'B', 3rd day 'A'.....so on. Then in how many days the work would be completed?

Assignment

- A** $15 \frac{5}{8}$ days
- B** $30 \frac{5}{8}$ days
- C** 31
- D** 16



2 mins Summary



Topic

Counting Theory, Time and Work

(Basic)

Alternate days



THANK - YOU